

Experiment-1:Regular Expressions

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 2
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/1.py
Phone Number: 9876543210
Email: student@gmail.com
Pattern matched successfully
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Experiment-2:Finite State Automaton

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/3.py
ab -> Accepted
aab -> Accepted
abab -> Accepted
abc -> Rejected
baa -> Rejected
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Experiment-3:Morphological Analysis Using NLTK

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/5.py
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\K.RAMESH\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data] C:\Users\K.RAMESH\AppData\Roaming\nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
cats -> cat
dogs -> dog
running -> running
played -> played
studies -> study
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Experiment-4:Finite-State Machine

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/1.py
cat -> cats
dog -> dogs
box -> boxes
bus -> buses
baby -> babies
church -> churches
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Experiment-5:Porter Stemmer

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/1.py
playing -> play
played -> play
plays -> play
studies -> studi
studying -> studi
connected -> connect
connection -> connect
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Experiment-6: Bigram N-Gram Model

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/1.py
Bigrams:
I -> ['like', 'like', 'like']
like -> ['Python', 'programming', 'machine']
Python -> ['I']
programming -> ['I']
machine -> ['learning']

Generated Text:
I like machine learning
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Experiment-7: POS Tagging + Stochastic POS Tagging

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/1.py
[nltk_data] C:\Users\K.RAMESH\AppData\Roaming\nltk_data...
[nltk_data] Unzipping tokenizers\punkt_tab.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] C:\Users\K.RAMESH\AppData\Roaming\nltk_data...
[nltk_data] Unzipping taggers\averaged_perceptron_tagger_eng.zip.
POS Tags:
The -> DT
student -> NN
reads -> VBZ
a -> DT
book -> NN
. -> .
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Experiment-8: Stochastic POS Tagging Using a Probabilistic Model

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/1.py
Stochastic POS Tagging:
the -> DT
student -> NN
reads -> VB
a -> DT
book -> NN
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```